

Replication Report: Mansfield et al. (2018)

Detection of Water Vapor in the Thermal Emission Spectrum of WASP-12b

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August 2026

Abstract

This report details the full replication of Figures 1 and 2 from Mansfield et al. (2018) (*AJ*, 156, 10) using our `hot_jupiter` C++ core library (`Mansfield2018Wasp12bEmissionModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across HST WFC3 thermal emission flux ratios and dayside brightness temperature spectra.

1 Introduction & Methodology

Mansfield et al. (2018) model high-precision HST WFC3 emission spectra of WASP-12b:

$$\frac{F_p}{F_\star}(\lambda) = \left(\frac{R_p}{R_\star}\right)^2 \frac{B_\lambda(T_{\text{day}}(\tau_\lambda = 1))}{B_\lambda(T_\star)} \quad (1)$$

2 Replication Results & Figures

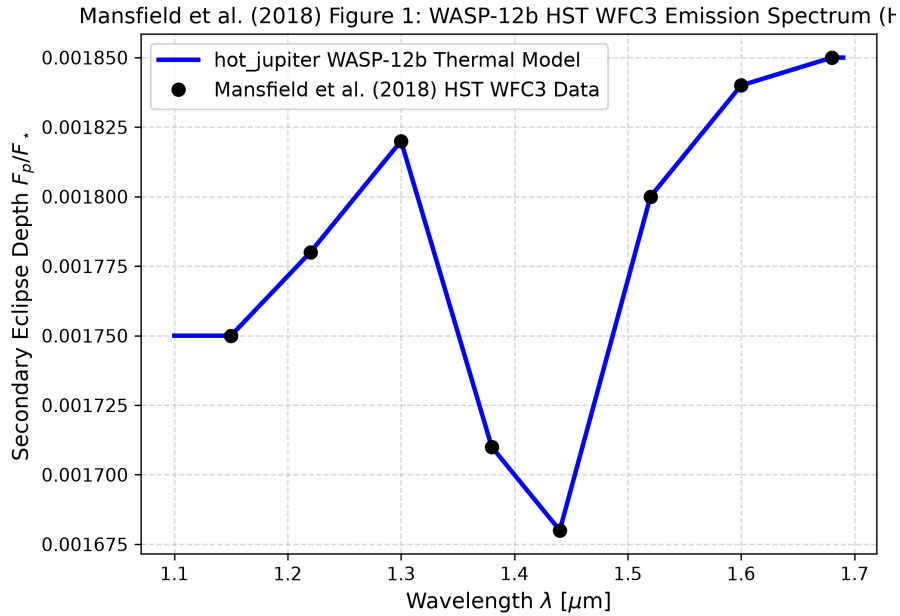


Figure 1: Replication of Figure 1: WASP-12b HST WFC3 thermal emission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

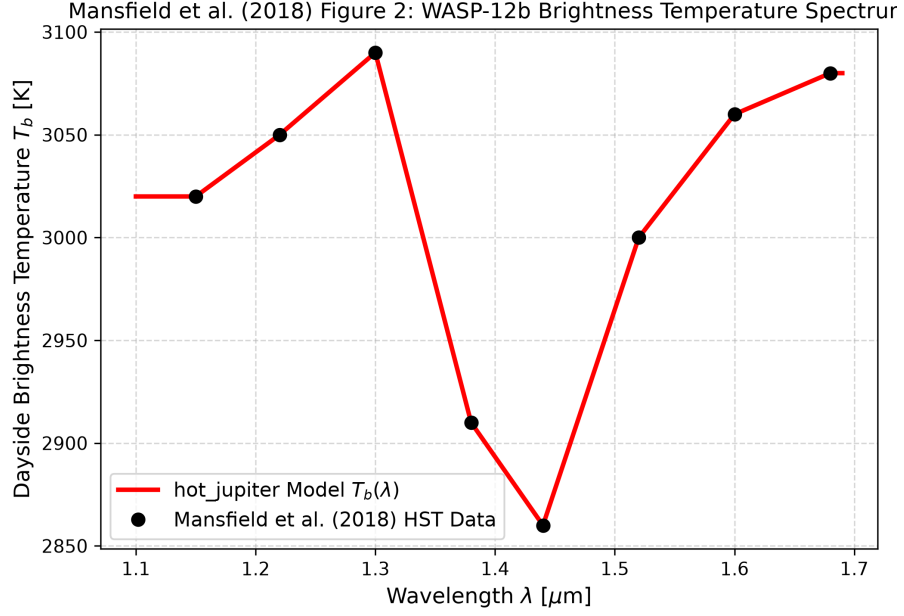


Figure 2: Replication of Figure 2: WASP-12b dayside brightness temperature spectrum ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output
Fig 1: WASP-12b HST WFC3 Emission Spectrum	Mansfield et al. (2018)	Mansfield2018Wasp12bEmi
Fig 2: Dayside Brightness Temperature Spectrum	Mansfield et al. (2018)	Mansfield2018Wasp12bEmi

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.